Cambridge Assessment International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*3537537102*

**BIOLOGY** **5090/22**

Paper 2 Theory **May/June 2022**

**1 hour 45 minutes**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Section A: answer **all** questions.
- Section B: answer **all** questions.
- Section C: answer **either** Question 8 **or** Question 9.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

**INFORMATION**

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.


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## Section A

Answer **all** questions in this section.

Write your answers in the spaces provided.

**1** The diagram shows the human brain and the table lists the functions of some parts of the human brain.



| part of brain | function |
|---|---|
| **A** | memory storage and control of conscious behaviour |
| **B** | control of body temperature |
| **C** | control of balance and coordination |
| **D** | control of unconscious activities such as breathing |

**(a)** Label on the diagram the parts of the brain identified in the table, using label lines and the letters **A** to **D only**. [4]

**(b)** **(i)** Name **one** structure in the skin that receives impulses from part **B** of the brain to control body temperature.

..... [1]

**(ii)** Outline how the action of this structure results in the control of body temperature.

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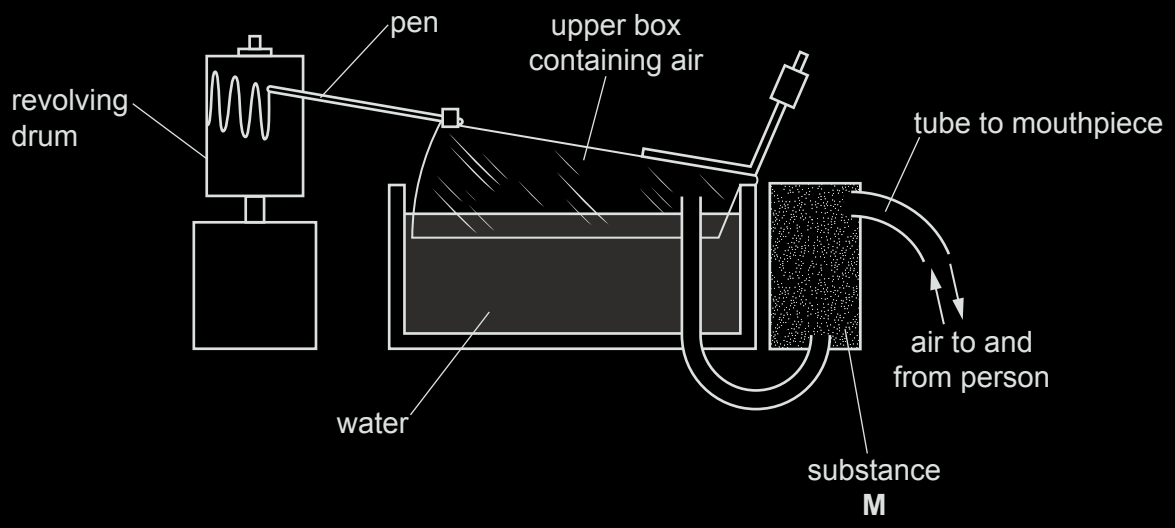
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..... [3]

[Total: 8]

**2** The diagram shows apparatus used to study breathing.




**(a) (i)** Describe what will happen to the upper box containing air as a person takes **one** breath **in** from the apparatus.

..... [1]

**(ii)** State the percentage of oxygen in atmospheric air.

..... %

Explain how the percentage of **oxygen** in the air in the upper box will change as a person breathes in **and** out through the apparatus several times.

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..... [3]

**(iii)** Substance **M** is present in the apparatus to absorb an excretory product from expired air.

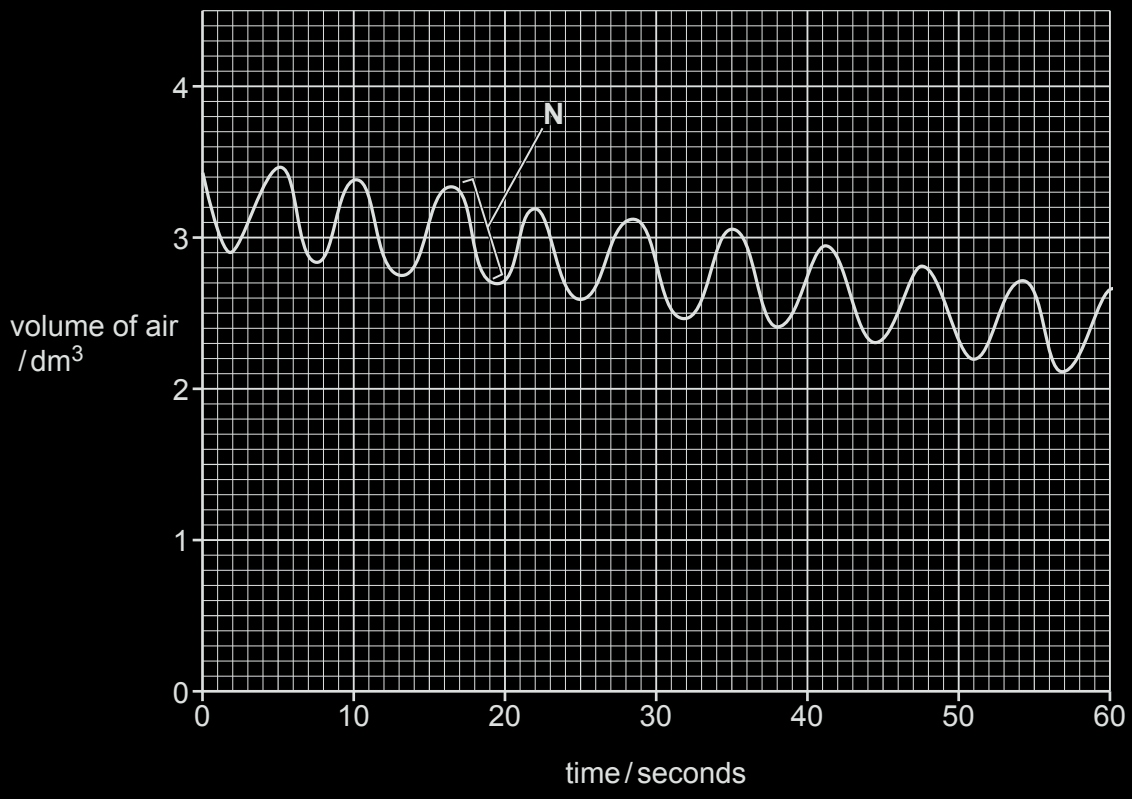
Name this excretory product.

..... [1]

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**(b)** The diagram shows a chart recorded by the pen on the revolving drum.

This chart was recorded for a period of one minute by a person at rest.




**(i)** Explain how the action of one **named** muscle caused the part of the chart labelled **N**.

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..... [3]

**(ii)** A second chart was recorded for a period of one minute by the same person after vigorous exercise.

State how an increased **rate** of breathing after exercise would be shown on this second chart.

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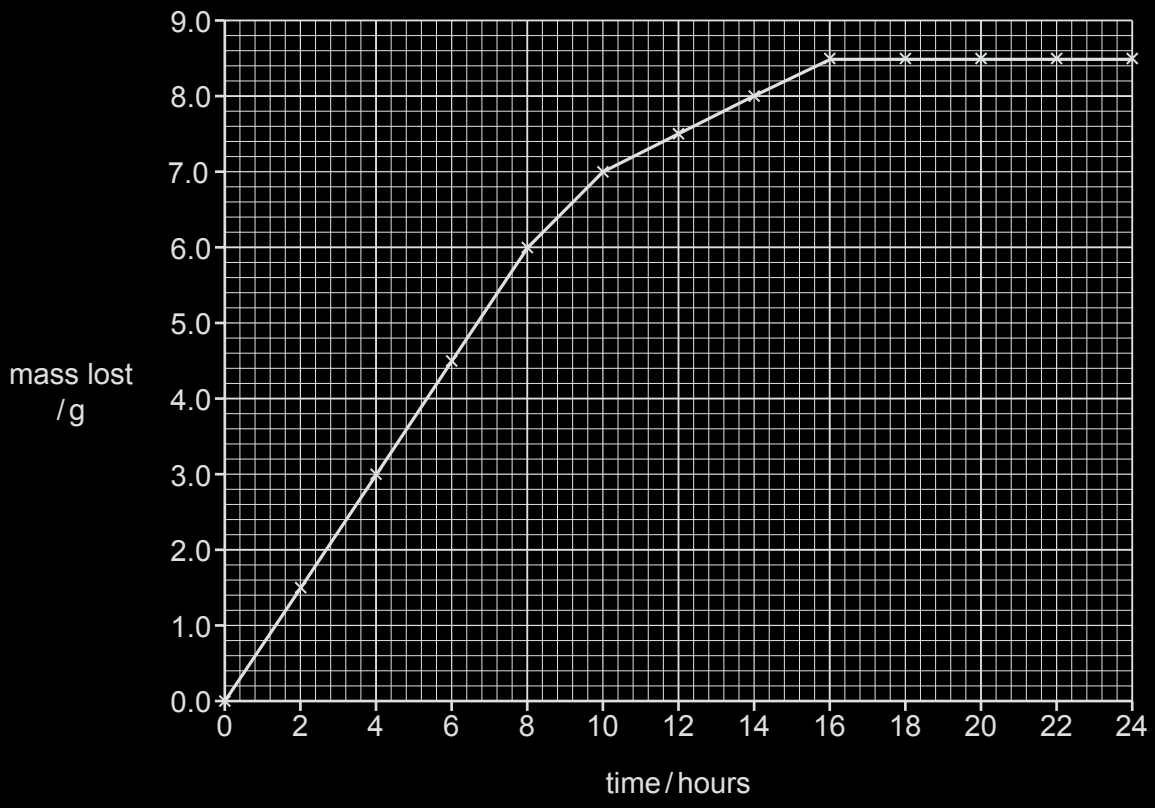
..... [1]

[Total: 9]

**3** A student carried out two investigations into the action of yeast.

**(a)** In the first investigation, the student dissolved 20.0 g of glucose in 100 $cm^3$ of water in a beaker. The student then added 3.5 g of yeast and used a balance to measure the loss in mass from the beaker over the next 24 hours.

The results of this investigation are shown in the graph.




**(i)** Calculate the rate at which mass was lost during the first 6 hours.

Space for working.

..... g per hour [2]

**(ii)** Explain why mass was lost during the investigation.

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..... [3]

**(iii)** The student then repeated this first investigation changing **only** the mass of yeast.

Draw a line on the graph on page 6 to show the pattern of results you would expect the student to obtain when using 7.0 g of yeast. [2]

**(b)** In a second investigation, the student prepared three samples of bread dough.

Each sample of dough had a volume of 50 $cm^3$ and contained:

- 1 g of yeast
- 25 $cm^3$ of water
- 40 g of flour.

**(i)** Starch molecules in the flour are used to provide the yeast with a source of glucose.

Describe how the action of a **named** chemical, produced by yeast cells, makes glucose available from starch.

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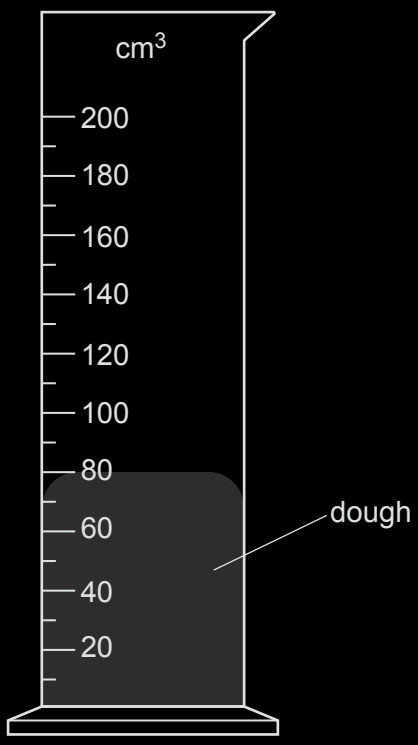
..... [2]

Each sample of dough was placed at the bottom of a 200 $cm^3$ measuring cylinder.

The three measuring cylinders were then placed in water-baths at different temperatures for 60 minutes. The temperatures chosen were 20 °C, 40 °C and 80 °C.

After this time, the student measured the volume of dough in each measuring cylinder.

The diagram shows the volume of dough in the measuring cylinder from the water-bath at a temperature of 20 °C at the end of the investigation.




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**(ii)** Estimate the volumes of the dough at the end of the investigation in the other two measuring cylinders, from water-baths at temperatures of:

40 °C ..... $cm^3$

80 °C ..... $cm^3$

[2]

**(iii)** Explain the results of this investigation.

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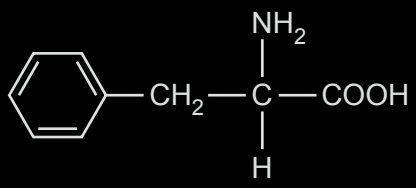
..... [3]

[Total: 14]

**4** There are many different types of amino acid.

One of these is phenylalanine.

The diagram shows the structure of a molecule of phenylalanine.




**(a)** Phenylalanine is metabolised by an enzyme with the name PAH.

**(i)** Name the human organ that metabolises amino acids such as phenylalanine.

..... [1]

**(ii)** Explain how the shape of a PAH enzyme molecule is important to enable phenylalanine to be metabolised.

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..... [3]

**(b)** A rare gene mutation reduces the activity of the PAH enzyme. The mutation results in an increased concentration of phenylalanine in the blood. This condition can cause damage to the brain.

**(i)** Describe what is meant by the term *gene mutation*.

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..... [2]

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**(ii)** Explain how a person with the condition should change their diet to reduce the risk of damage to the brain.

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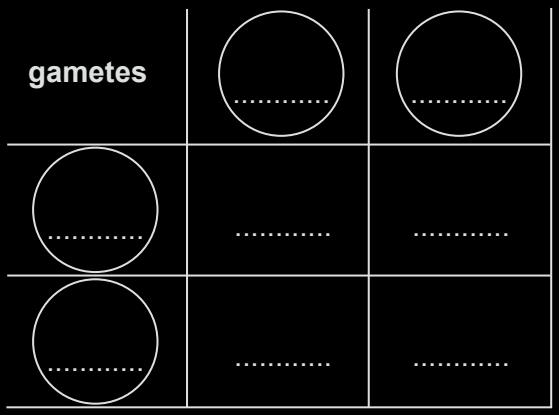
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**(iii)** The gene mutation produces a recessive allele, **h**. A person with a heterozygous genotype is described as a carrier. A carrier will **not** show symptoms of the condition.

Complete the genetic diagram to show the probability of two carriers reproducing to have offspring with the condition.



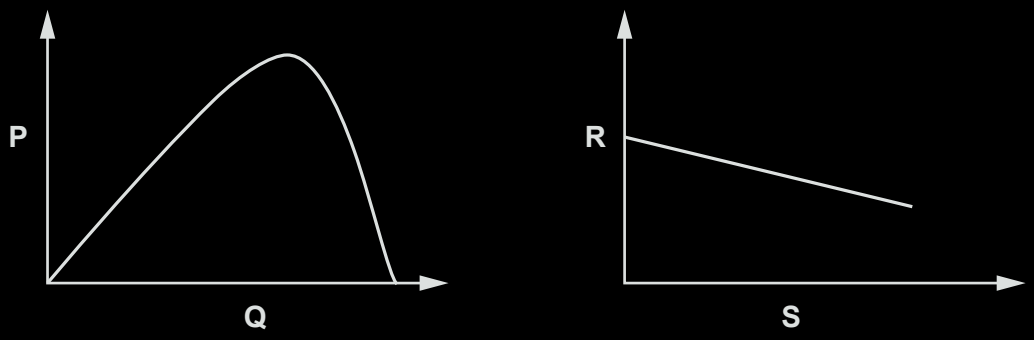

probability of offspring with the condition .....
[3]

[Total: 12]

**5** **(a)** Biological processes in plants are affected by environmental factors.

The graphs each show the effect of changing one environmental factor on a biological process.

The labels on the axes for each graph have been replaced with the letters **P**, **Q**, **R** and **S**.




Identify the correct axis label for each of **P**, **Q**, **R** and **S**.

Select from the following possible labels.

**temperature** **carbon dioxide concentration** **rate of transpiration**

**light intensity** **rate of photosynthesis** **air humidity**

**P** .....

**Q** .....

**R** .....

**S** .....

[4]

**(b)** Describe the process of transpiration in the leaf of a plant.

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..... [3]

[Total: 7]

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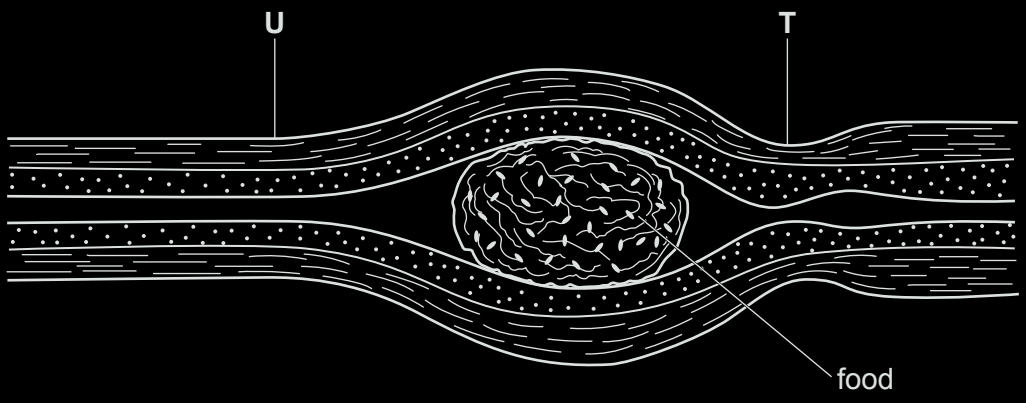
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## Section B

Answer **both** questions in this section.

Write your answers in the spaces provided.

**6** The diagram shows food inside part of the small intestine.




**(a)** Describe and explain how food is moved between the two points, **T** and **U**, labelled on the diagram.

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..... [7]

**(b)** State, for a normal healthy movement of food through the small intestine,

**one** necessary component of a balanced diet

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**one** food that is a good source of this component

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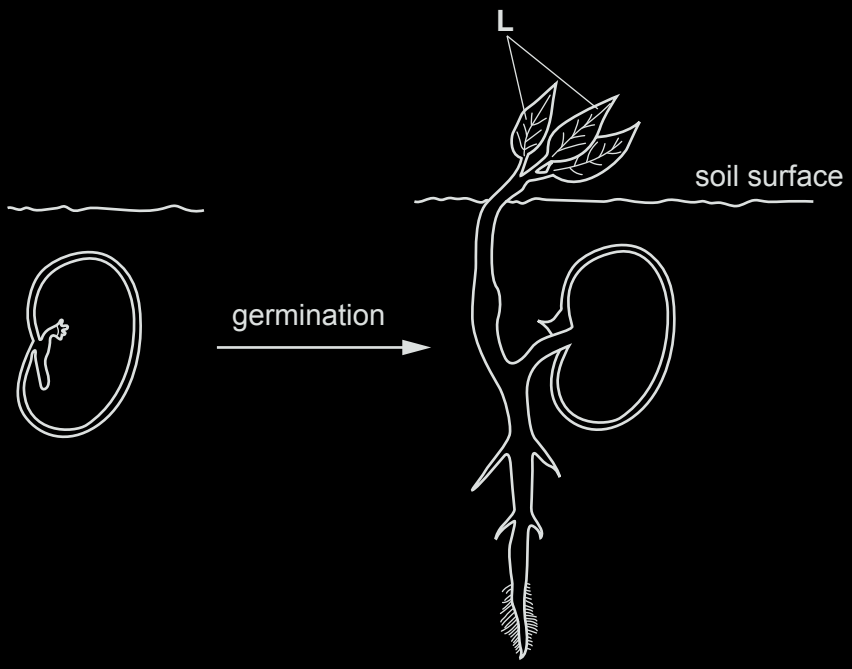
**one** effect of malnutrition resulting from a lack of this component.

.....

[3]

[Total: 10]

**7** The diagram shows a cross-section through a seed before germination and the same seed after it has become a seedling.




**(a)** Outline the importance of the following in seed germination:

enzymes .....

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cell division .....

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[7]

**(b)** Explain why it is important that the structures labelled **L** in the seedling are above the soil surface.

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..... [3]

[Total: 10]

## Section C

Answer **either** Question 8 **or** Question 9.

Write your answers in the spaces provided.

**8** **(a)** Compare the pulmonary artery and the vena cava in terms of their structure and function.

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..... [6]

**(b)** Explain the role of platelets and the importance of this role.

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..... [4]

[Total: 10]

9 **(a)** Compare the ureter and the urethra in terms of their functions in adult humans.

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..... [4]

**(b)** Explain how the process of dialysis maintains the correct chemical composition of the blood for a person with kidney disease.

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..... [6]

[Total: 10]

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